Final Report – Week 4

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Domain: Security Operations Center (SOC) Operations & Threat Detection
Platform Stack: Wazuh 4.x, Windows Server 2022, Kali Linux, Ubuntu Server, Sysmon, Velociraptor, VirusTotal

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1. Introduction

This report summarizes Week 4 SOC practical activities covering threat hunting, SOAR simulation, post-incident analysis, adversary emulation, evidence handling, metrics evaluation, and a full capstone incident response lifecycle.
The objective was to demonstrate structured detection engineering, automation logic, incident handling, forensic validation, and executive-level reporting within a controlled lab environment.

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2. Lab Environment

Component	Role
Ubuntu Server	Wazuh Manager (SIEM)
Windows Server 2022	Monitored Endpoint
Kali Linux	Attacker System
Windows 11	Management Host
Velociraptor	Endpoint Forensics
Sysmon	Advanced Process Logging

Logs collected:

- Windows Security Event Logs
- Sysmon Process Creation Logs
- Network Connection Artifacts

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3. Task 1 – Threat Hunting Practice

Unauthorized Authentication Attempts

Hunting Hypothesis:
An attacker is attempting unauthorized access using invalid credentials (MITRE ATT&CK T1078 – Valid Accounts).

Detection Query

win.system.eventID:4625

Observations

Timestamp	Event ID	Target User	Source IP	Logon Type
19:31:08	4625	FakeUser	192.168.52.137	3

Multiple repeated failures from the same IP confirmed brute-force/credential testing behavior.

Threat Intelligence Validation

- IP checked via VirusTotal
- No public reputation (internal lab IP)
- Behavioral risk remained high

Outcome
Hypothesis confirmed. Unauthorized authentication attempts were detected successfully.

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4. Task 2 – SOAR Playbook Simulation

Automated Brute Force Containment
Trigger Rule: Wazuh Rule 60204 – Multiple Logon Failures

Playbook Flow

1. Receive alert
2. Extract source IP
3. Reputation lookup
4. Block IP via firewall
5. Document incident

Containment Command

sudo iptables -A INPUT -s 192.168.52.137 -j DROP

Execution Summary

Step	Status

Alert Triggered	Success
IP Extracted	Success
Reputation Check	Completed
Firewall Block	Success

Result: Automated logic successfully simulated structured response workflow.

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5. Task 3 – Post-Incident Analysis

Brute Force Login Incident

Root Cause Analysis (5 Whys)

Primary Root Cause:

RDP exposed without account lockout and security baseline enforcement.

SOC Metrics

Metric	Value
MTTD	3 minutes
MTTR	12 minutes
Dwell Time	15 minutes

Lessons Learned

- Implement account lockout policy
- Enforce MFA for RDP
- Apply segmentation
- Automate containment

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6. Task 4 – Alert Triage with Automation

Suspicious File Download Investigation

Alert: DNS Event ID 1014 during large file download

Triage Actions

- Extracted SHA256
- Submitted to VirusTotal
- Detection ratio: 0/70

Playbook Simulation

Step	Result
Hash Extraction	Success
Reputation Check	0/70
Case Creation	Completed
Closure	Benign

Conclusion: False alarm confirmed; structured triage process validated.

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7. Task 5 – Evidence Analysis

Tool Used: Velociraptor
Artifact: *Linux.Network.Netstat*

Findings
- Active outbound Firefox TCP sessions
- No suspicious unauthorized processes

Integrity Verification
SHA256:
4d65d1dac15d7ad02635cd09031408617f94c2d4197ec334bc531fbaaa789de7

Chain of Custody

Item	Description	Date	Hash
TCP4.csv	Netstat Output	2026-02-25	Verified

Evidence integrity preserved successfully.

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8. Task 6 – Adversary Emulation

Scenario
Simulated phishing execution using encoded PowerShell.

Techniques Simulated
- T1566 – Phishing
- T1059.001 – PowerShell
- T1087 – Account Discovery

Detection Query
rule.mitre.id:T1059.001

Observations

Timestamp	Technique	Status
15:44:11	T1059.001	Detected
15:44:10	T1087	Detected

Gaps Identified

- No email-stage monitoring
- No automated containment
- No alert-to-ticket integration

Result: Strong execution-stage visibility, improvement required for automation.

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9. Task 7 – Security Metrics & Executive Reporting

Authentication Attack Simulation (WinRM)

Metric	Value
MTTD	~0–5 sec
False Positive Rate	0%
Dwell Time	0

Interpretation

- Near real-time detection capability
- No successful compromise
- Manual containment needs automation improvement

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10. Task 8 – Capstone Project

Comprehensive SOC Incident Lifecycle

Attack

RDP brute-force attempt using invalid credentials.

Command:

xfreerdp3 /v:192.168.52.129 /u:administrator /p:wrongpass123

Detection

Rule ID: 60122
Event ID: 4625
Technique Mapping:

- T1110 – Brute Force
- T1021 – Remote Services

Containment
PowerShell Firewall Rule:

New-NetFirewallRule -DisplayName "Block-Kali-Attacker" -Direction Inbound -RemoteAddress 192.168.52.137 -Action Block

SOC Metrics

Metric	Value
MTTD	1 minute
MTTR	24 minutes
Dwell Time	25 minutes

Root Cause
Remote access hardening not enforced.

Recommendations
- Restrict RDP access
- Enable MFA
- Configure lockout policies
- Automate IP blocking
- Conduct periodic access audits

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11. Overall Outcomes

This internship cycle demonstrated:
- Hypothesis-driven threat hunting
- Real-time brute-force detection
- Simulated SOAR automation workflows
- Structured post-incident analysis
- Forensic evidence collection and validation
- Adversary emulation aligned with MITRE ATT&CK
- SOC metric calculation and executive-level reporting
- End-to-end incident lifecycle handling

The SOC lab environment successfully validated detection visibility, response coordination, and measurable performance evaluation.

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12. Conclusion

Week 4 successfully demonstrated operational SOC maturity across detection, triage, containment, forensic validation, adversary emulation, and reporting.

Detection capability is strong and real-time.

Response processes are structured but require deeper automation for reduced MTTR.

Hardening improvements and proactive control enforcement will further enhance defensive posture.

The capstone confirms readiness to manage authentication-based threats within a structured SOC workflow environment.